7 **(a)** An electron is confined inside a one-dimensional potential well of length 0.50 nm with infinitely high potential barriers at $x = 0$ and $x = 0.50$ nm.

Within the well, the potential energy $U(x) = 0$, and outside the well, $U(x) = \infty$.

(i) By considering the de Broglie wavelength of the electron trapped in this well, explain why the electron has specific allowed energy levels.

.....

.....

.....

.....

.....[2]

(ii) The electron emits a single photon when it transitions from the second excited state to the ground state.

Determine the wavelength of the photon.

wavelength = ... m [3]

(iii) Explain, with reference to the Heisenberg Uncertainty Principle, why the ground state energy of the electron cannot be zero.

.....

.....

.....

.....[2]

(b) An electron bound within a hydrogen atom has allowed energy levels shown in Fig. 7.1.

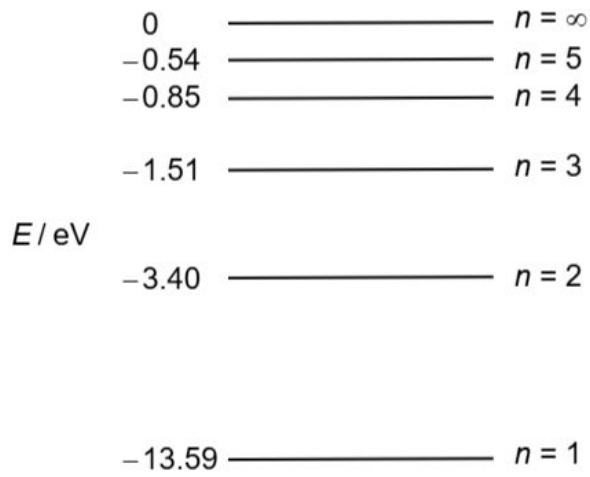


Fig. 7.1

(i) Suggest one reason why the energy levels of this electron are different from those of the electron in **(a)**.

.....

.....

.....[1]

(ii) The electron emits a single photon when it transitions from the second excited state to the ground state. Determine the wavelength of the photon.

wavelength = ... m [2]

[Total: 10]